

Solving the Sensor-Based Activity Recognition Problem (SOAR): Self-Supervised, Multi-Modal Recognition of Activities from Wearable Sensors

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ABSTRACT

Feature extraction lies at the core of Human Activity Recognition (HAR): the automated inference of what activity is being performed. Traditionally, the HAR community used statistical metrics and distribution-based representations to summarize the movement present in windows of sensor data into feature vectors. More recently, learned representations have been used successfully in lieu of such handcrafted and manually engineered features. In particular, the community has shown substantial interest in self-supervised methods, which leverage large-scale unlabeled data to first learn useful representations that are subsequently fine-tuned to the target applications. In this tutorial, we focus on representations for single-sensor and multi-modal setups, and go beyond the current de facto of learning representations. We also discuss the economic use of existing representations, specifically via transfer learning and domain adaptation. The proposed tutorial will introduce state-of-the-art methods for representation learning in HAR, and provide a forum for researchers from mobile and ubiquitous computing to not only discuss the current state of the field but to also chart future directions for the field itself, including answering what it would take to finally solve the activity recognition problem.

CCS CONCEPTS

• **Human-centered computing** → **Ubiquitous and mobile computing**; **Empirical studies in ubiquitous and mobile computing**; • **Computing methodologies** → *Machine learning*.

KEYWORDS

human activity recognition, representation learning, self-supervision, multi-modal learning

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1 INTENDED AUDIENCE

This is an introductory to intermediate tutorial. Participants will include UbiComp and HCI researchers, industry practitioners, and students. We will expect participants to have familiarity with the basics of machine learning and Python coding. Ideal participants would have experience with coding in notebooks such as Jupyter and Google Colab.

2 PREVIOUS OFFERINGS OF THE TUTORIAL

The organizers ran related tutorials at previous conferences, which the proposed tutorial builds upon. At UbiComp 2015 Ploetz (along with Anind Dey [then CMU, now UW], Mayank Goel [now CMU], Nils Hammerla [then Newcastle University]) organized a full day tutorial on “Machine Learning for UbiComp”, which was very well attended (sold out on first day of registration). Ploetz (together with Nic Lane [then Oxford and Nokia Bell Labs, now Cambridge and Samsung] and Sourav Bhattacharya [then Nokia Bell Labs, now Samsung] and Yu Guan [Newcastle University]) then organized tutorials on “Deep Learning for UbiComp”, which were held at MobiCase 2016 and UbiComp 2017. Both tutorials were also well attended (sold out).

3 MAIN CONTACT PERSON

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4 ORGANIZERS

Harish Haresamudram: Harish Haresamudram is a fourth year PhD student with the School of Electrical and Computer Engineering at Georgia Institute of Technology. His research involves understanding human movements and behaviors via the opportunistic use of body worn sensing platforms from wearable devices,

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as they enable truly in-the-wild data collection. He is particularly interested in learning representations via techniques that require minimal supervision, by leveraging large quantities of unlabeled movement data.

Chi Ian Tang: Chi Ian Tang is a final-year PhD student in Computer Science at the University of Cambridge. His research focuses on developing scalable and data-efficient solutions for mobile sensing, with a special interest in leveraging massive amounts of unlabeled sensing data collected in the wild, to develop more generalizable models for mobile sensing in order to provide deeper insights about user behaviors. His work has appeared in ICML, UbiComp (ACM IMWUT), and ICASSP, among others, and he has been a reviewer for IMWUT, ISWC and ACM HEALTH.

Sungho Suh: He is a Senior Researcher at the German Research Center for Artificial Intelligence (DFKI) in Germany since 2020. He received the Ph.D. degree in Computer Science at the Technische Universität Kaiserslautern, Germany in 2021, and the B.S. and M.S. degrees from the School of Electrical Engineering and Computer Science, Seoul National University, Seoul, South Korea, in 2009 and 2011, respectively. His research interests are machine learning algorithms, such as sensor data processing, computer vision, multi-modal processing, and generative models, with a focus on industrial applications.

Paul Lukowicz: Paul Lukowicz is both Scientific Director at the German Research Center for Artificial Intelligence (DFKI GmbH) and Professor of Computer Science at RPTU Kaiserslautern-Landau in Germany since 2012 where he heads the Embedded Intelligence group. His research focuses on context-aware ubiquitous and wearable systems including sensing, pattern recognition, system architectures, models of large-scale self-organized systems, and applications in areas ranging from healthcare through industry 4.0 to smart cities.

Thomas Ploetz: Thomas Ploetz is a computer scientist with expertise and almost 20 years of experience in Pattern Recognition and Machine Learning research (Ph.D. from Bielefeld University, Germany). His research agenda focuses on applied machine learning, that is developing systems and innovative sensor data analysis methods for real world applications. Primary application domain for his work is computational behavior analysis where he develops methods for automated and objective behavior assessments in naturalistic environments. Main driving functions for his work are "in the wild" deployments and as such the development of systems and methods that have a real impact on peoples' lives. Thomas is currently Associate Professor at the School of Interactive Computing at Georgia Institute of Technology.

5 MOTIVATION / RELEVANCE TO THE COMMUNITY

The advent of Deep Learning (DL) based methods in the general machine learning community was followed by its adoption in the UbiComp community, leveraging the presence of annotated datasets to learn features that are useful for recognizing activities [3, 8]. However, existing works have demonstrated that the collection of large-scale labeled data for mobile sensing tasks is particularly difficult [1, 7, 9], due to the difficulty in interpreting sensor streams (such as accelerometer traces). Data collection efforts are limited

to small-scale studies requiring a high level of participation from subjects. This denies the ability to develop truly deep networks (of hundreds of layers) for wearables-based applications. Despite this, their effectiveness has rendered them the gold standard for activity recognition and speaks to their widespread usage. While there is a paucity of large-scale labeled data, simply collecting large quantities of sensor data without annotations is straightforward due to the ubiquitous nature of the devices. More recently, a new class of methods has seen increasing interest, which leverages this inherent disparity between the apparent ease of data collection and the more challenging nature of data annotation, by learning representations on unlabeled data, before fine-tuning the features to target applications. This family of approaches, which are based on 'Self-Supervised Learning', has heralded a paradigm change within the community, with a gold rush of new methods aiming to make economic use of the publicly available unlabeled datasets for learning effective representations [2, 4–6, 10, 11].

In this tutorial, we will present the progress of wearable sensor-based representation learning from handcrafted heuristics to the current state-of-the-field comprising self-supervised learning. Subsequently, we will also present how available learned weights (i.e., the representations) can be leveraged economically for improved downstream recognition, via transfer learning and domain adaptation, while also highlighting how the availability of more sensing modalities could be useful. We will have a hands-on session using Colab notebooks, which will involve important components of human activity recognition and popular methods for tackling the recognition problem. Further, we will have an open discussion that will discuss, debate, and (potentially) chart the direction of the field and solicit opinions on what it would take to finally solve the activity recognition problem. Put together, the goal of the tutorial is to bring researchers together to present the progress of the recognition problem, and also to derive possible future directions for tackling it.

6 BRIEF OUTLINE OF THE TOPICS TO BE COVERED

- Overview of established practices and challenges in building ML models for sensor data analysis, including the Activity Recognition Chain (ARC).
- Conventional machine learning vs. deep learning for sensory data analysis.
- Unsupervised and self-supervised learning methods for wearables-based human activity recognition.
- Economic use of learned weights via transfer learning and domain adaptation.
- Use of other available modalities through multi-modal learning.

7 FORMAT AND DETAILED SCHEDULE

The tutorial is planned as a half-day event that focuses on three main components:

- (1) Presentations by the organizers.
- (2) Hand-on exercises.
- (3) Open discussion regarding the methods, the field, and its future.

Table 1: Tentative schedule for the tutorial.

Time	Block	Presenter
8:30 AM - 9:00 AM	Registration and Setup	All
9:00 AM - 9:15 AM	Start: Welcome, Introduction, Overview of HAR and classical machine learning techniques.	Thomas/Paul
9:15 AM - 10:00 AM	Introduction to self-supervised learning, early approaches for wearable sensor data.	Harish/Sungho/Ian
10:00 AM - 10:30 AM	Contrastive learning approaches in general, and for sensor data; leveraging multiple sensors.	Harish/Sungho/Ian
10:30 AM - 11:00 AM	Overflow time and brief break	–
11:00 AM - 11:30 AM	Generating sensor data: augmentation, GANs, diffusion, ‘virtual’ IMU data from videos.	Harish/Sungho/Ian
11:30 AM - 12:15 PM	Exploring various representations via Colab notebooks.	Harish/Sungho/Ian
12:15 PM - 12:30 PM	General discussion about the state-of-the-field; audience thoughts on future methods; Conclusion.	Thomas/Paul

The detailed (tentative) breakdown of the tutorial can be found in Tab. 1.

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